

Nicolas Rubiano

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EDUCATION

University of British Columbia

Bachelor of Science, Computer Science | GPA: 3.7/4.0

Vancouver, BC

Sep 2022 – Nov 2025

EXPERIENCE

Software Engineer

Sep 2025 – Present

Inversiones Bloom

Delaware, USA (Remote)

- Designed an AWS SQS-based queue system with DynamoDB conditional writes to eliminate failed investment transactions during high-demand campaign closings, recovering the 15% of investors who previously abandoned after a failed attempt, verified to handle 1,000 concurrent investors with zero transaction loss.
- Wrote automated tests for critical concurrency and edge-case scenarios in the transaction processing system, raising coverage from 50% to 85%+ and catching 15+ defects in CI before deployment (reducing production incidents by 25% in the following month).
- Built fraud detection into the authentication system (geolocation and impossible-travel checks with rate limiting) using Claude Code to cut implementation time roughly in half while holding validation under 2.5ms and blocking 20+ suspicious logins monthly.
- Created the project's first backend documentation (Lambda architecture and DynamoDB access-pattern design), validated through a team walkthrough, cutting new-engineer ramp-up time from over a month to under a week.

Teaching Assistant

Jul 2025 – Aug 2025

University of British Columbia

Vancouver, BC

- Restored automated Python grading scripts for 7 labs within a week after they stopped functioning, clearing a grading backlog affecting 35+ students and re-enabling reliable evaluation via GitHub Actions.
- Mentored 15+ students through office hours across 4 Go-based distributed systems labs, helping identify and correct real concurrency, fault-tolerance, and scalability issues in their implementations.

Data Engineer

Jan 2022 – April 2022

Movii

Victoria, Canada (Remote)

- Built an automated data pipeline to clean, validate, and upload government subsidy datasets from Excel to Redshift, reducing marketing campaign turnaround time from about a week to two days and eliminating 20+ hours of manual data entry monthly.
- Optimized Redshift queries powering payment-tracking dashboards (reducing retrieval time by 20% and query costs by 10% across hundreds of thousands of records), enabling responsive, cost-efficient reporting for stakeholder analysis.
- Built Amazon QuickSight dashboards flagging payment gaps, enabling marketing to identify and follow up with 25+ unpaid subsidy recipients per campaign, totaling \$7,500+.

PROJECTS

MangaSearch

Jan 2025 – Present

- Built a local manga quote search engine using Go and Python (EasyOCR), letting users instantly find any quote's exact chapter and page across their entire collection with no cloud dependency, achieving 16 pages/min across 4 concurrent workers with 0% data loss on transient errors.
- Delivered full-text fuzzy search via Elasticsearch and Redis caching, achieving 1-2ms warm queries versus 62ms cold, a 30x speedup on repeat searches, with a Cobra CLI for searching and managing the engine.
- Designed a file change detection system using a HashMap diff algorithm to avoid redundant OCR processing on every 30-minute rescan, achieving 100% page skip on unchanged files with 0 duplicate jobs entering the queue.

Pairly

Jan 2026 – Present

- Built a real-time collaborative code editor in Go, where multiple users edit simultaneously, horizontally scalable via WebSocket pods and Redis pub/sub, verified at 0% error rate across 44 concurrent users on 3 replicas.
- Implemented version history for documents using Kafka and MongoDB, achieving 90% cache hit rate and full history replay averaging 0.8s across a 30-day edit log.
- Deployed on Kubernetes with independent auto-scaling per service (3-10 server pods, 2-6 executor pods), 25s/30s graceful shutdown, distributed tracing via OpenTelemetry and Jaeger, and CI/CD pipeline via GitHub Actions.

TECHNICAL SKILLS

Languages: Go, Java, Python, SQL, TypeScript

Infrastructure: Kubernetes, Docker, Kafka, RabbitMQ, Redis

Databases: PostgreSQL, MongoDB, Elasticsearch, DynamoDB

Cloud & Tools: AWS, GitHub Actions, OpenTelemetry, Prometheus

Frameworks: Spring Boot, Gin, FastAPI, React

AI Dev Tools: Claude Code, GitHub Copilot, GPT-4o